

Nuclear Fuel Performance

NE-533
Spring 2026

UO₂ RADIATION EFFECTS

Radiation Damage

- The macroscopic and observable results of exposure of solids to energetic particles are collectively known as radiation effects
- Heat production from the nuclear fuel to generate electricity ensues mostly from the slow down of the fission products by nuclear or electronic interactions
- As a direct consequence, there are also defects created along the path of the fission fragments leading to modification of the physical properties of the fuel
- Most Frenkel defects (interstitial-vacancy pair) are produced by secondary collision cascades
- The dominant defect type is anion Frenkel pairs
- Local charge neutrality prevents/inhibits the formation of interstitials and vacancies as independent processes, as the cations and anions are charged species

To balance charge, how many oxygen vacancies need to be created for every uranium vacancy?

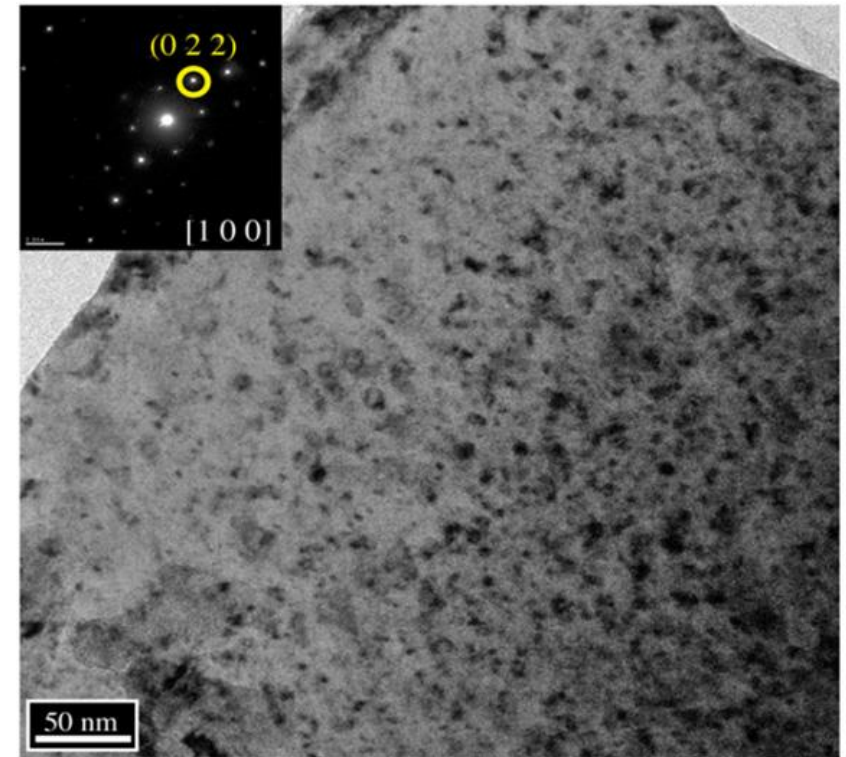
Two

U+4, O-2

This forms a Schottky trio defect

Loop Formation

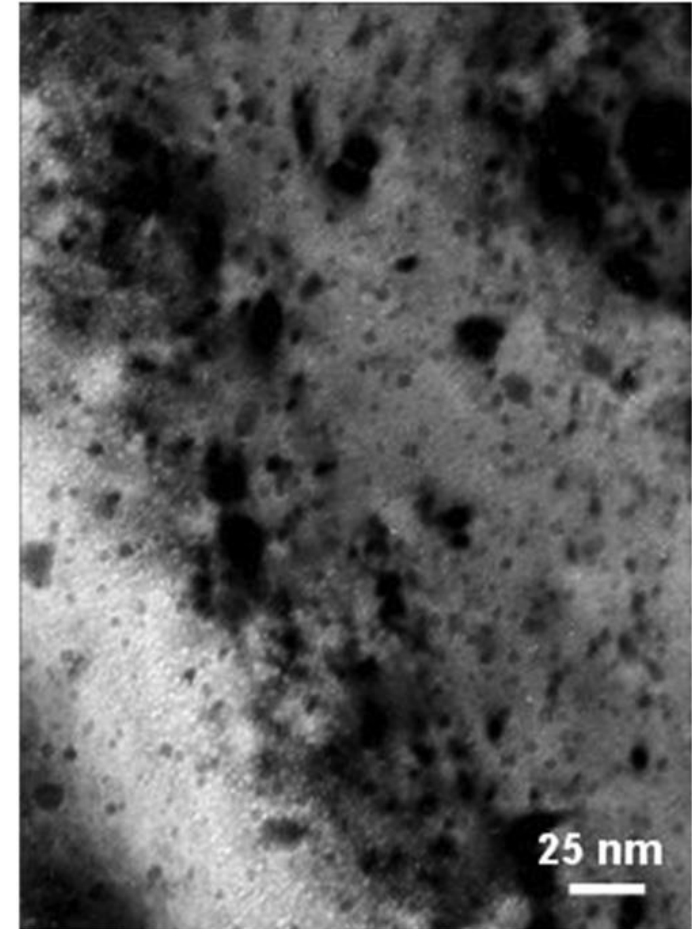
- Dislocations are generally a mixture of screw and edge components in the shape of a loop
- The most important slip system in UO₂ is the one which minimizes the electrostatic energy, the easy slip along $\{100\}\langle 110\rangle$ tends to minimize the repulsive force between cations
- Prismatic loops are of interstitial type and consist of a disc-shaped layer of defects in a cluster
- This defect cluster is more energetically favorable than dispersed defects in the bulk
- Vacancies will cluster and form into voids



Prismatic loops in UO₂ from alpha particle irradiation

Voids

- Due to preferential absorption of interstitials at dislocations, we are left with a slight excess of vacancies in the bulk to nucleate and grow voids
- Void nucleation can occur homogeneously by the collection of vacancies in the matrix, or heterogeneously at structural features (precipitates, grain boundaries, etc.)
- The vacancy-rich radiation damage core is considered to be an important void nucleation mechanism in UO₂
- Vacancy diffusion then allows void growth
- Gas atoms can often serve to stabilize small voids



Voids (bright spots) from alpha irradiation
in UO₂

UO₂ Radiation Effects

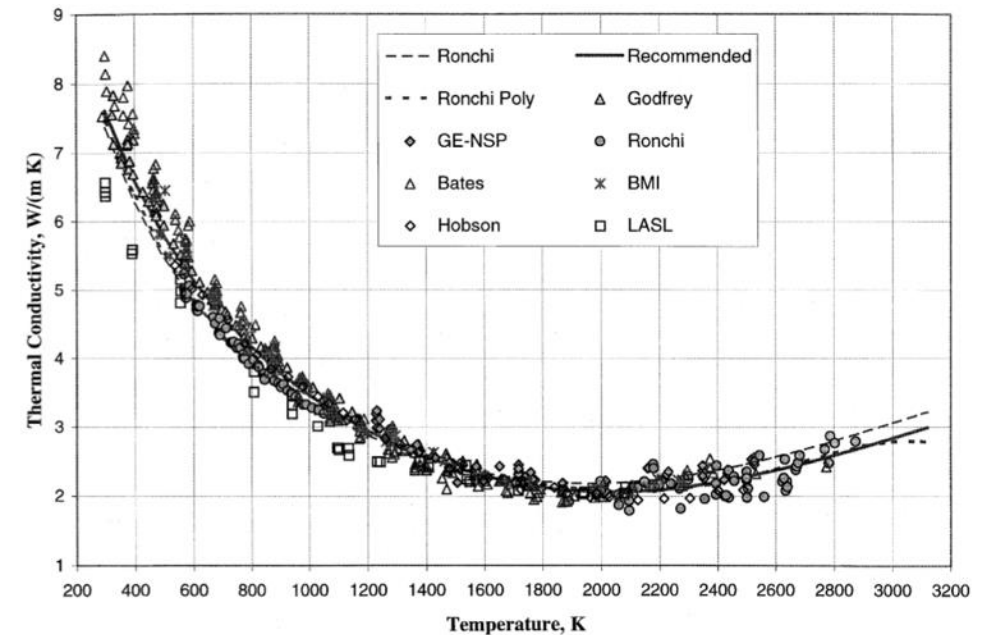
- Chemistry changes
 - two fission products are generated from each fission, resulting up to about 3% of the chemical species present at the end of life for commercial fuel
 - the amount of fission products change both chemical and physical properties
- Physical property changes
 - The physical effects caused by atomic displacements are rather complex and depend on the relative sink strengths of a given material for interstitials and vacancies and on the temperature
 - lattice parameter, thermal conductivity, elasticity, formation of the high burnup structure

Thermal Conductivity Degradation

- One of the major effects of defect formation is the degradation of thermal diffusivity, due to phonon scattering
- Results obtained on irradiated UO₂ show that the thermal conductivity is decreasing with increasing burn-up, and that at equal burn-up, samples with higher irradiation temperatures have higher thermal conductivity
- There are mesoscopic and atomistic features that affect the thermal conductivity
- The porosity existing in fresh fuels evolves during irradiation and has a strong impact on heat transport, because voids/bubbles have significantly lower k_{th} than the lattice
- Precipitates are formed by fission products that are insoluble in the UO₂ lattice, forming oxide or metallic inclusions: can use composite theory to get effective thermal conductivities; typically, these precipitates **INCREASE** the thermal conductivity of the fuel

Thermal Conductivity Degradation

- Actinides, rare earths, and transition metals can form mixed oxides with UO_2 (soluble fission products): these atoms act as phonon scattering centers as a result of the differences in bonding, radii, or mass, and thus decrease k_{th}
- Volatile fission products/gases are partially in solution and partially in bubbles, where they act as phonon scatterers or as low k_{th} composite fractions
- Point defects and extended defects contribute to the degradation of the thermal conductivity by scattering or limiting the mean free path of the phonons

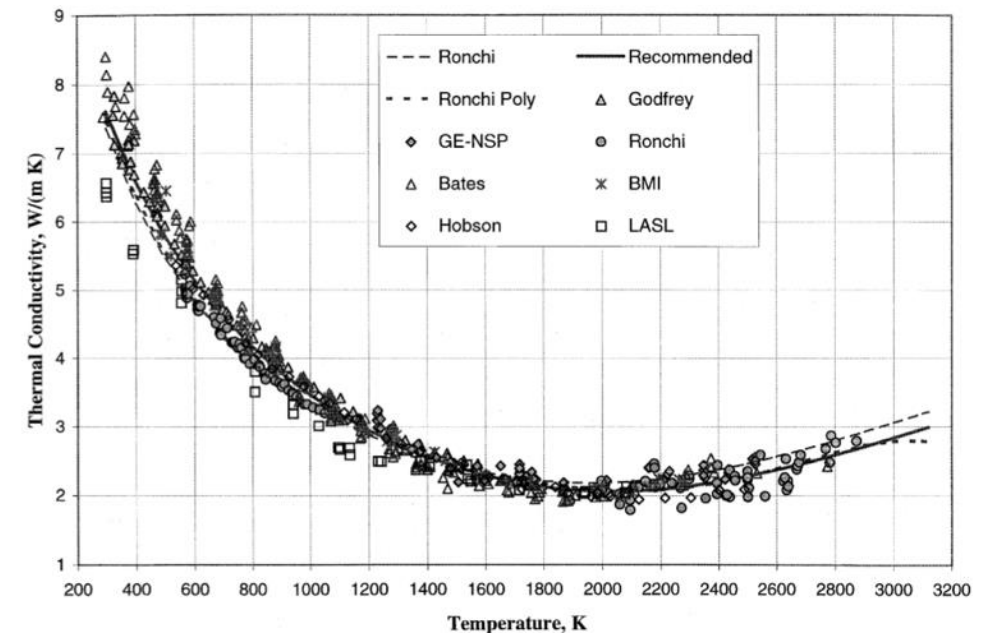


UO₂ Thermal Conductivity

- We have established correlations for thermal conductivity of fresh UO₂ fuel as a function of temperature

$$k_0 = \frac{100}{7.5408 + 17.629t + 3.6142t^2} + \frac{6400}{t^{5/2}} \exp\left(\frac{-16.35}{t}\right)$$

- Where $t = T/1000$
- The first part of the equation describes the phonon interactions
- The second part describes electronic transport which becomes significant at high temperature



UO₂ Thermal Conductivity

- UO₂ thermal conductivity is low and decreases more during reactor operation
- The thermal conductivity has been collected after various amounts of burnup to make empirical fits

$$\lambda_{95} = \frac{1}{A(x) + aG + B(x)T + f_1(Bu) + f_2(Bu)g(Bu)h(T)} + \frac{C}{T^2} \exp\left(\frac{-D}{T}\right)$$

